

Yiming Tang

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EDUCATION

University of Southern California

08/2022 - 05/2024

Master of Science, Computer Science, GPA: 3.7/4.0

Coursework: Machine Learning, Deep Learning, Natural Language Processing, Multi-modality

University of California, Irvine

09/2017 - 12/2021

Bachelor of Science, Computer Science/Business Information Management, GPA: 3.8/4.0

Coursework: Algorithm, Data Management, Game Development, Software Development, Intro to Artificial Intelligence, Computer System, Information Retrieval

RESEARCH EXPERIENCE

A Unified Model for Multi-Party Social Signal Processing with the [Prof. Jesse Thomason](#) at USC 05/2023 - 01/2025

- Developing a unified transformer model for predicting multi-party social signals to improve feeding people with disabilities.
- Designed and trained a novel causal masking strategy for a **GPT-2**-based transformer to handle multi-party social signals.

LLM for Search Engine Optimization with [Prof. Yue Zhao](#) and [Prof. Xiyang Hu](#) at USC 11/2024 - 05/2025

- Develop an interpretable attack prompt for LLM search engine optimization, which ranks the target product at first.
- Utilize **COLD-Decoding** to generate a more stealth and readable prompt.

Modality-agnostic Fusion for Emotion Recognition with [Prof. Mohammad Soleymani](#) at USC 01/2024 - 05/2024

- Designed a parameter-efficient shared encoder for emotion recognition that processes features individually for multi-tasking (classification and reconstruction), achieving comparable performance with fewer parameters.
- Improved emotion recognition performance over unimodal approaches by implementing early, late, and gated fusion across unimodal models

Sequential Data Analysis with [Prof. Pavlos Protopapa](#) at Harvard 09/2020 - 12/2020

- Conducted sentiment analysis, named entity recognition, and machine translation from scratch by constructing models using **TensorFlow** framework, including **RNN**, **LSTM**, **GRU**, encoder-decoder **transformer**, and **GRU** cell.
- Compared the performance on each task with various model architectures by adding attention layers, integrating **distilBERT**, and improving recurrent unit cells.
- Reproduced the results of [MultiFiT](#) on efficient multi-lingual language model fine-tuning using TensorFlow

PUBLICATIONS

- [1] [M3PT: A transformer for multimodal, multi-party social signal prediction with person-aware blockwise attention](#)
Yiming Tang, Abrar Anwar, Jesse Thomason
Workshop on International Conference on Robotics and Automation (ICRA 2025) **Best Paper**
- [2] [StealthRank: Stealthy Product Ranking Manipulation in LLM-based Recommendations](#)
Yiming Tang, Yi Fan, Chenxiao Yu, Tiankai Yang, Yue Zhao, Xiyang Hu
Submitted to Empirical Methods in Natural Language Processing (EMNLP 2025)
- [3] [Humor detection in edited news headlines: predicting humor ratings for news headlines with atomic change](#)
Yiming Tang
International Conference on Electronic Information Engineering and Computer Communication (EIECC 2021)

WORK EXPERIENCE

Chongqing Zenith Information Technology Co., Ltd 06/2021 - 07/2021

Full Stack Developer

- Developed a web application for data management using **Spring Boot** and **Javascript**, **HTML**, **VUE**.
- Designed and implemented the backend architecture for application via **Spring MVC**
- Created **REST** controllers in **Spring MVC** to handle HTTP requests (GET, PUT, POST, DELETE), facilitating data exchanges and responses to frontend requests
- Integrated **Spring Data JPA** for efficient data handling between **MySQL** and applications

PROJECTS

Prompting with Attention 01/2023 - 05/2023

- Improved zero-shot **RoBERTa** classification performance by using **p-tuning** to derive a base continuous prompt, then ensembling over the logits of prompt variations with an attention layer. Uses less parameters compared to fine-tuning.

AI Text Detection 01/2023 - 05/2023

- Utilized **GPT-2** to generate fake text paired with human-written text, modified the objective function with a triplet loss, and employed **RoBERTa** to achieve over 90% accuracy in distinguishing between human-written and AI-generated text on story topics.

SKILLS

Programming Languages: Python, C++, C, Java

Frameworks: numpy, pandas, sklearn, pytorch, pytorch-lightning, tensorflow